

锁控板规格书

Net (Lan converter) Lock board Specification

产品概述 Products description

该锁控板采用有线以太网通讯接口，支持多种电控锁，支持锁信号反馈，支持 12V 和 24V 的电控锁，且锁反馈方式可通过软件配置，具备独立 PTC 自恢复保险丝，支持锁短路保护，保护电流为 3A，提供配置工具软件，用户可自由更改通讯协议、和锁反馈类型。

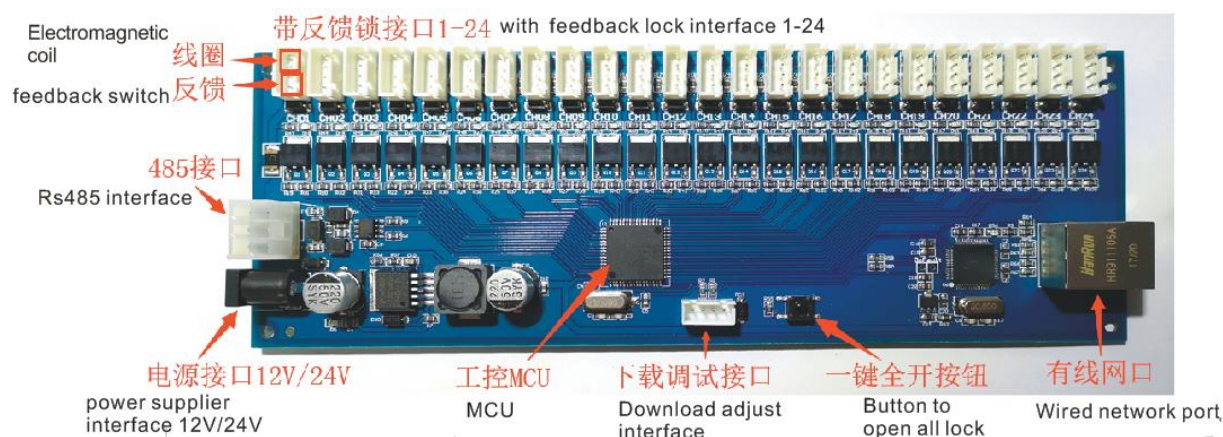
The lock control board adopts wired Ethernet communication interface, supports a variety of electric control locks, supports lock signal feedback, supports 12V and 24V electric control locks, and the lock feedback method can be configured by software, with independent PTC resettable fuses, and supports locks. Short-circuit protection, protection current is 3A, configuration tool software is provided, users can freely change the communication protocol and lock feedback type.

应用领域 Application field

该锁控板广泛应用于快递柜，自动售货机，储物柜，自动化药房，共享图书柜，生鲜柜，共享餐柜等等。

The lock control board is widely used in express cabinets, vending machines, storage cabinets, automated pharmacies, shared book cabinets, fresh food cabinets, shared dining cabinets and so on.

硬件说明 Hardware interface description



产品特点

- 支持 10/100M，全双工/半双工自适应以太网接口，兼容 802.3 协议
- 支持 MDI/MDIX 线路自动转换
- 支持 DHCP 自动获取 IP 地址，支持 DNS 域名访问
- 通过上位机软件、串口命令设置芯片工作模式、端口、IP 等网络参数
- 工作模式支持 TCP CLIENT、TCP SERVER 和 UDP CLIENT、UDP SERVER 4 种模式
- 支持并提供虚拟串口软件

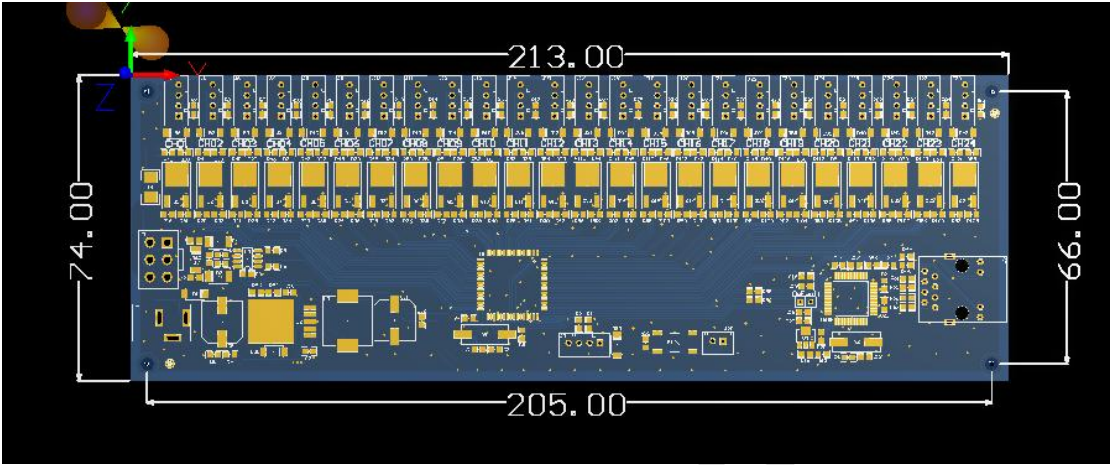
Feature of product

- Support 10 / 100M, full-duplex / half-duplex adaptive Ethernet interface, compatible with 802.3 protocol
- Support MDI / MDIX line automatic conversion
- Support DHCP to obtain IP address automatically, support DNS domain name access
- Set the chip's working mode, port, IP and other network parameters through host computer software and serial commands
- Working mode supports TCP CLIENT, TCP SERVER and UDP CLIENT, UDP SERVER 4 modes
- Support and provide virtual serial port software

电气参数 Electrical parameters

工作电压 Working voltage	DC6-35V
待机功耗 Standby power consumption	Less 1W
工作温度 Working temperature	-30℃~ +80℃
通讯接口 Communication interface	TCP.UDP
485 级联数量 485 connect quantity	Max 32 pcs
驱动锁电流 drive the lock current	Max 3A
读锁反馈 read lock status	Support
开锁输出时间 open lock time	200ms
程序升级接口 program update interface	TTL series /Typc C
看门狗 watchdog timer	Support

一、机械尺寸 Size (mm)



通讯协议 Protocol

1、概述 description

本协议采用 TCP 通信方式，板子作 TCP 客户端，板子自带 485 扩展接口，同一个 485 网络里，最多可扩展 32 个 485 锁控板。

This protocol uses TCP communication method. The board is used as a TCP client. The board comes with a 485 expansion interface. In the same 485 network, it can expand up to 32pcs 485 lock control boards.

2、数据帧格式 Date frame format

名称 name	起始符 start character	帧长度 Frame length	板地址 Board address	指令字 Instruction word	数据域 Data field	校验 Verify
长度 (字节)Length (bytes)	4	1	1	1	n	1

起始符：四字节，默认为“WKLY”，用户可通过配置工具修改

帧长度：一个字节，为从起始到校验字节的字节数，取值范围为 0x08 ~ 0xFF。

板地址：一个字节，对于一般指令，只有当地址和从机自身地址相同时，才执行指令。

指令字：一个字节，从机回复时，命令字相同

数据域：长度由命令字决定。对于命令应答，模块返回数据域的第一个字节为状态字节；如果命令执行错误，此数据域中只包含状态字节。状态字节 0x00 表示执行正确，其它数值表示执行错误。

校验字节：从帧起始符到数据域最后一个字节逐字节的异或（XOR）值。

Start character: 4 bytes, default to "WKLY", user can use the CONFIG TOOL amend

Frame length: 1 byte, is the byte from the start to verify, the range is 0x08 ~ 0xFF.

Board address, 1 byte, for normal command, only when the location address is same the board address, the the command works

Instruction word: 1 byte, respond from the board, the command byte is same

Date field: the length is based on the command word. For respond of command, the first byte is the status byte when the module back to date field; for example, if the command execution error, the date field just include the status byte. **The status bytes 0x00 means ok, otherwise, are wrong**

Verify byte: from the frame start character to the last byte and gradually byte date of date field (XOR)

指令列表 Command list

命 令 字 Command word	说明 Description	备注 Remark
0x80	设备发送心跳包 device send the heartbeat packet	Only when the board as client-side
0x81	设备向服务器注册 device registers to the server	Only when the board as client-side
0x82	开锁 open lock	
0x83	读门状态 check the door status	
0x84	查询所有状态 check all status	
0x85	主动上传门状态变化 active to post the door status	
0x86	开全部锁 open all locks	
0x87	开多个通道锁 open some channel locks	
0x88	通道常开 channel open	
0x89	通道关闭 channel close	
0x94	多个通道输出延时关闭 Multiple channel output delay shutdown	

3、 数据帧详解 Date frame details

3.1 心跳包 Heartbeat packet

设备端发送：设备端发送： device transmitter sending

起始符 start character	帧 长 度 Frame length	板 号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0x10	0x00	0x80	设 备 ID 号 device ID (8)	XOR

服务器回复 Server response:

起始符 start character	帧 长 度 Frame length	板 号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0x09	0x00	0x80	status (1)	XOR

Demo: 设备端向服务器发送心跳命令：

设备端发送： 57 4B 4C 59 10 00 80 30 30 30 30 30 30 31 98

服务器回复： 57 4B 4C 59 09 00 80 00 80

注：服务器没有给设备端发送数据的时间超过 30 秒，设备端将主动主动上传心跳包，以维持 TCP 连接。服务器需要对心跳包进行回复，如果连续 4 个心跳包没有回复，设备端将断开 TCP 连接，进行重连。

Demo : device transmitter send the heartbeat command to Server

Device transmitter send: : 57 4B 4C 59 10 00 80 30 30 30 30 30 30 30 31 98

Server response: 57 4B 4C 59 09 00 80 00 80

Note: if the server donot send the date to device transmitter more than 30s, the device transmitter will post the heartbeat packet active, to maintain the TCP connection. Server need to respond for heartbeat packet. If there is 4 heartbeat packet no respond, the device transmitter will cut off the TCP connection, then re-connect.

3.2 设备向服务器注册 Device register to server

设备连上服务器后，将主动向服务器发送注册命令

When the device connect to the service, then send the register command to server active.

设备端发送： Device transmitter sending:

起始符 start character	帧长度 Frame length	板号 board number	指令字 Instruction word	数据域 Data field	校验 Verify
0x57 0x4B 0x4C 0x59	0x26	0x00	0x81	Device ID No. (8)+ Device type (2) +CCID (20)	XOR

设备 ID 号：8 个字节，可通过配置软件对设备端进行配置。

设备类型：两个字节，用于服务器区分中断设备，00 12 代表 12 路锁柜板

CCID：板子上 SIM 卡的 CCID

Device ID number: 8 bytes, can through the configuration tool to configure the device transmitter

Device type: 2 bytes, for server area to cut off the device. 00 12 is for 12 way lock cabinet board

CCID: CCID of SIM card from board

服务器回复 Server respond:

起始符 start character	帧长度 Frame length	板号 board number	指令字 Instruction word	数据域 Data field	校验 Verify
0x57 0x4B 0x4C 0x59	0x09	0x00	0x81	status (1)	XOR

状态：注册成功为 0x00；注册失败为 0xFF，设备将自动断开 TCP 连接

Demo: 设备端向服务器发送注册命令：

设备端发送：57 4B 4C 59 26 00 81 30 30 30 30 30 30 31 00 25 38 39 38 36 30 34 33
32 30 31 31 38 43 30 32 30 33 37 30 37 FA

服务器回复：57 4B 4C 59 09 00 81 00 81

Status: register success show: 0x00. faild show 0xFF, and board cut off the TCP connection by itself

Demo : device transmitter send the register command to Server

Device transmitter sending: 57 4B 4C 59 26 00 81 30 30 30 30 30 30 31 00 25 38 39
38 36 30 34 33 32 30 31 31 38 43 30 32 30 33 37 30 37 FA

Server response: 57 4B 4C 59 09 00 81 00 81

3.3 开单个通道锁 Open single channel lock

服务器发送：Sever sending

起 始 符 start character	帧长度 Frame length	板号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0xXX	0x00	0x82	Channel number (1)+order number(n)	XOR

设备端回复: Device transmitter respond

起 始 符 start character	帧长度 Frame length	板 号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0xXX	0x00	0x82	Status (1)+Channel number (1)+ order number(n)	XOR

订单号: 锁控板只对订单号原样返回, 长度为 0-24, 如果不需要, 也可以为空

锁状态: 0x00 代表门打开, 0x01 代表门关闭, 0xFF 代表开锁越界

Demo: 开 0 号板的通道 1:

服务器发送: 57 4B 4C 59 09 00 82 01 83

设备端回复: 57 4B 4C 59 0B 00 82 00 01 00 81

Order number : lock control board just back for order original sample , length 0-24, if donot need, can be empty

Lock status: 0x00 means door open, 0x01 means door close, 0xFF means lock cross the border

Demo: open the channel 1 of number 0 board

Server sending: 57 4B 4C 59 09 00 82 01 83

Device transmitter respond: 57 4B 4C 59 0B 00 82 00 01 00 81

3.4 读单个锁状态 Read single lock status

服务器发送: Sever sending:

起始符 start character	帧 长 度 Frame length	板 号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0x09	0x00	0x83	channel (1)	XOR

设备端回复: Device transmitter respond

起始符 start character	帧 长 度 Frame length	板 号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0x0B	0x00	0x83	Status(1)+channel number(1)+ lock status(1)	XOR

Demo: 读取 0 号板的通道 1 状态:

3.6 Posting the door status actively

设备端发送: Device sending

起始符 start character	帧 长 度 Frame length	板 号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0x0A	0x00	0x85	Channel number (1)+lock status(1)	XOR

锁状态: 0x00 代表门打开, 0x01 代表门关闭, 手动关门与通过非指令手段开门, 锁控板才会上传该指令。

Demo: 0 号板的通道 1 门关闭

设备端发送: 57 4B 4C 59 0A 00 85 01 01 86

服务器不需要回复

Lock status: 0x00 for door open, 0x01 for door close, close the door by hands
And through non-command way open door, then the lock control board post the Command.

Demo: channel 1 door close of number 0 board

Device sending: 57 4B 4C 59 0A 00 85 01 01 86

Server donot need to respond

3.7 Open all channel locks

服务器发送: Server sending

起始符 start character	帧 长 度 Frame length	板 号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0x08	0x00	0x86	none	XOR

设备端回复 Device transmitter respond:

起始符 start character	帧 长 度 Frame length	板 号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0x09	0x00	0x86	Status (1)	XOR

状态: 0x00 代表成功

Demo: 打开 0 号板的全部通道:

服务器发送: 57 4B 4C 59 08 00 86 87

设备端回复: 57 4B 4C 59 09 00 86 00 86

Status: 0x00 for succeed

Demo: open all channels of number 0 board

Sever sending: : 57 4B 4C 59 08 00 86 87

Device transmitter respond: 57 4B 4C 59 09 00 86 00 86

3.8 开多个通道锁 Open few channel locks

服务器发送: Sever sending

起始符 start character	帧 长 度 Frame length	板号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0xXX	0x00	0x87	Channel quantity(1) +channel number(n)	XOR

设备端回复 Device transmitter respond:

起始符 start character	帧 长 度 Frame length	板号 board number	指 令 字 Instruction word	数据域 Data field	校 验 Verify
0x57 0x4B 0x4C 0x59	0x09	0x00	0x87	Status (1)	XOR

状态: 0x00 代表成功

Demo: 打开 0 号板的 1、2、3 通道

服务器发送: 57 4B 4C 59 0C 00 87 03 01 02 03 81

设备端回复: 57 4B 4C 59 09 00 87 00 87

Status: 0x00 for succeed

Demo: open the channel 1. 2. 3 of number 0 board

Sever sending: : 57 4B 4C 59 0C 00 87 03 01 02 03 81

Device transmitter respond: : 57 4B 4C 59 09 00 87 00 87

3.9 通道常开 Keep channel open

该命令用于控制继电器、照明灯等可以持续通电的设备，当锁接口接瞬时通电开锁的电控锁时，执行该命令可能会烧坏电控锁。

This command is used to control relays, lights, etc., which can be continuously power on. When the lock interface is connected to the electronically controlled lock which is

ready to open, executing this command may burn the electronically controlled lock

服务器发送: Sever sending:

起始符 start character	帧长度 Frame length	板号 board number	指令字 Instruction word	数据域 Data field	校验 Verify
0x57 0x4B 0x4C 0x59	0x09	0x00	0x88	Channel number(1)	XOR

设备端回复: Device transmitter respond

起始符 start character	帧长度 Frame length	板号 board number	指令字 Instruction word	数据域 Data field	校验 Verify
0x57 0x4B 0x4C 0x59	0x0A	0x00	0x88	Status(1)+Channel number(1)	XOR

状态: 0x00 代表成功

Demo: 0 号板的 1 通道持续开启

服务器发送: 57 4B 4C 59 09 00 88 01 89

设备端回复: 57 4B 4C 59 0A 00 88 00 01 8A

Status: 0x00 for succeed

Demo: channel 1 keeps open of number 0 board

Sever sending: 57 4B 4C 59 09 00 88 01 89

Device transmitter respond: 57 4B 4C 59 0A 00 88 00 01 8A

3.10 通道关闭 Channel close

服务器发送: Sever sending

起始符 start character	帧长度 Frame length	板号 board number	指令字 Instruction word	数据域 Data field	校验 Verify
0x57 0x4B 0x4C 0x59	0x09	0x00	0x89	Channel number(1)	XOR

设备端回复: Device transmitter respond:

起始符 start character	帧长度 Frame length	板号 board number	指令字 Instruction word	数据域 Data field	校验 Verify
0x57 0x4B 0x4C 0x59	0x0A	0x00	0x89	Status(1)+Channel number(1)	XOR

状态: 0x00 代表成功

Demo: 0 号板的 1 通道关闭

服务器发送: 57 4B 4C 59 09 00 89 01 88

设备端回复: 57 4B 4C 59 0A 00 89 00 01 8B

Status: 0x00 for succeed

Demo: channel 1 close of number 0 board

Sever sending: 57 4B 4C 59 09 00 89 01 88

Device transmitter respond: 57 4B 4C 59 0A 00 89 00 01 8B

3.11 校验计算详细说明 Verify Calculation details description

以下为主板 MCU 内部代码中的检验计算函数

Following is : interal code to verify the calculation function of Mother board MCU

/******

* 函数名 Function : ComputXor

* 描述 Description : 异或累加校验计算函数 XOR cumulative check calculation function

* 输入 Input : InData, 参与校验计算的数组 number of verify the calculation; Len 参数校验计算的数据长度 length of verify the calculation

* 返回 Back : 检验结果 Verity result

* 说明 Description : 无 nono

** 作 者 Writer: Vyoung

** 日 期 Date : 2015-9-15

*****/

uint8 ComputXor(uint8 *InData, uint16 Len)

```
{
    uint8 Sum = 0;
    uint16 i;
    for(i = 0; i < Len; i++)
    {
        Sum ^= InData[i];
    }
    return Sum;
}
```